

Flask-UI Design Decisions and User Flow

Design Decisions

The user interface (UI) was designed with the goals of clarity, usability, and responsiveness in mind.

1. Framework & Styling

- Flask (Jinja2) for server-side templating and dynamic rendering.
- Bootstrap 5 for responsive design, ensuring consistent styling across devices.
- Custom CSS for subtle branding (rounded cards, soft shadows, linear gradient background).

2. Layout Choices

- Two-column layout:
 - Left Column → Review input form, model selection, quick actions, and model status.
 - Right Column → Displays prediction results dynamically.
- Card components are used for grouping related UI elements (form, status, results) for better visual hierarchy.

3. Color & Typography

- Soft blue gradient background for a calm, professional appearance.
- Bootstrap's primary, success, and info badges to indicate model status and highlight prediction confidence.
- System-friendly font stack for fast rendering and cross-platform consistency.

4. Interactive Elements

- Dropdown for model selection (Model A = balanced dataset, Model B = imbalanced dataset).
- Quick action buttons pre-fill example reviews for faster testing.
- Progress bar visually represents prediction confidence percentage.

5. Error & Feedback Handling

- Flask's flash() messages notify the user of invalid input or missing models.
- Clear placeholder and example texts guide the user to enter meaningful reviews.

User Flow

The flow focuses on making predictions with minimal steps while allowing quick experimentation.

1. Landing on Home Page (/)

- User sees the app title, model status, and review input area.
- UI indicates which models are loaded and ready.

2. Entering Review

- User types or pastes a product review in the text area.
- Optional: Click a “Quick Action” button to auto-fill a sample review.

3. Choosing Model

- User selects either **Model A** (balanced LSTM) or **Model B** (imbalanced LSTM) from the dropdown.

4. Submitting

- Clicking **Predict** sends the review and model choice to /predict via POST request.
- Server:
 - Cleans and validates the review text.
 - Runs the review through the selected model.
 - Returns the prediction label and confidence score.

5. Viewing Results

- The right panel updates with:
 - Predicted rating/label.
 - Confidence percentage (with progress bar).
 - Model used.
 - Tip/example note for better inputs.

6. Adjusting & Retesting

- User can:
 - Modify the review text.
 - Switch models for comparison.
 - Try quick examples again.
- No page reload needed beyond form submission.